



Technical Note

Using undisturbed soil samples to study how rock fragments and soil macropores affect the hydraulic conductivity of forest stony soils: Some methodological aspects



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ABSTRACT

Forest stony soils have rock fragments and many macropores, the latter arising mainly from the activity of soil fauna and plant root decay. We still lack knowledge about the hydraulic properties of this type of soil, especially when considering the influence of both factors – rock fragments and macropores – on the saturated hydraulic conductivity. This research aims to supplement the method we proposed before to test the hydraulic conductivity of undisturbed soil samples (Ilek and Kucza, 2014; Kucza and Ilek, 2016). We propose to enhance the method by taking into account the impact of rock fragments and macropores on the hydraulic conductivity of forest stony soils, particularly on the hydraulic conductivity of the medium conducting water in the soil, that is, soil matrix. We studied, using artificial samples, how rock fragments influence hydraulic conductivity. To test the enhanced method, we used 33 undisturbed samples taken from the top layers (0–12 cm) of forest stony soils. The results show that rock fragments in the soil can both increase and decrease the infiltration of water by the soil matrix. Rock fragments with diameters close to 20 mm had the least influence on water infiltration. Soil macropores in the samples always increased water infiltration. Empty macropores increased it from two to twenty-four times while filled macropores from three to five times. Our method gives the opportunity to study the hydraulic conductivity of macropores and their physical and chemical properties.

1. Introduction

Saturated hydraulic conductivity affects surface runoff and water infiltration through the soil profile. Thus, hydraulic conductivity is one of the most important hydraulic properties of soil – but at the same time, it is one of the most difficult ones to measure (Jačka et al., 2014). Many indirect and direct methods for measuring hydraulic conductivity have been proposed for homogeneous soils (e.g., Ankeny et al., 1991; Bagarello et al., 2004; Meadows et al., 2005; Schindler and Müller, 2006; Ghanbarian-Alavijeh et al., 2010; Di Prima, 2015; Di Prima et al., 2016), and different methods often give different values of hydraulic conductivity (Lee et al., 1985; Mohanty et al., 1994; Jačka et al., 2014).

Measuring hydraulic conductivity is difficult in forest stony soils, due to the presence of rock fragments and soil macropores in the soil (Hendrayanto et al., 1998; Ma et al., 2010). It is because of rock fragments that the hydraulic properties of stony soils differ from non-stony ones (Sharma et al., 1993). Depending on the type, fraction, size and

porosity, rock fragments can either enhance or worsen soil hydraulic and retention properties. Since rock fragments have in general a low ability to retain water, stony soils usually have low water-holding capacity (Hanson and Blevins, 1979; Cousin et al., 2003; Parajuli et al., 2017), thereby decreasing the volume of water available for plants (Moustakas et al., 1995). Usually, they also hinder water infiltration, by reducing the cross-section through which water infiltrates down the soil profile (Ma and Shao, 2008; Novák and Křápa, 2012) and increasing the tortuosity of water flow paths (Childs and Flint, 1990; Zhou et al., 2011). Under certain conditions, however, rock fragments reduce surface runoff, increasing the infiltration rate by creating new pores at the boundary between rock fragments and soil matrix (Grant and Struchtemeyer, 1959; Jung, 1960; Poesen and Ingelmo-Sanchez, 1992; Tokunaga et al., 2003).

Rock fragments pose many problems in soil hydrology research, from sampling to measuring water content in soil by indirect methods (Coppola et al., 2013). It is likely for these difficulties that rock

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fragments and their influence on the water-content measurement are often ignored when describing the hydraulic properties of soils.

Water flows through the soil in principal through the soil matrix and macropores (Weiler and Naef, 2003) or other preferential pathways. Macropores are free spaces in the soil, created due to soil fauna activity, plant root decay, wetting and drying as well as freezing and thawing of the soil, and erosive processes (Beven and Germann, 1982). Many authors distinguish macropores from non-macropores based on commonly accepted diameter limits (e.g., Chen and Wagenet, 1992; Lin, 2010), and according to Greco (2002), the diameter of macropores is orders of magnitude larger than that of soil matrix textural pores. Vegetation strongly influences what types of macropores are formed in the soil. Shrub areas create large, open macropores leading to deep soil layers (Archer et al., 2002), and decaying grass roots form systems of macropores of even size (Beven and Germann, 1982). Petersson et al. (1987) showed a strong positive relationship between hydraulic conductivity and the root mass of trees. This relationship, however, was more apparent in older stands, with a greater share of dead roots and soil macropores formed after their death. The bark of tree roots usually decays longer than xylem does, so plant-derived macropores are often filled with material originating from the decay of the root itself (Beven and Germann, 1982). Such macropores can account for about 35% of the volume of forest soils, and the number of macropores decreases with the depth of the soil profile (Aubertin, 1971). Difficult quantification of macropores (Allaire et al., 2009; Dohnal et al., 2009) and large variability of macropore flow are among the reasons for hydrological rainfall-runoff models to seldom include water flow through macropores (Weiler and Naef, 2003; Vareecken et al., 2016). Although macropores account for a small share of the total soil porosity, they may disproportionately influence soil infiltration properties (Buttle and House, 1997), especially under heavy rainfall that exceeds the soil's ability to conduct water (Lee and Foster, 1991; Mando et al., 1996).

Methods of testing the hydraulic conductivity of forest stony soils and soils containing soil macropores are constantly being developed (Ma et al., 2010; Novák et al., 2011; Ilek and Kucza, 2014; Germer and Braun, 2015). Knowledge on hydraulic properties of such soils, however, is still limited, particularly in terms of the influence of rock fragments and soil macropores on hydraulic conductivity – the most important reason being a lack of simple methods for determining it in this type of soil. To estimate hydraulic conductivity, *in situ* tests make it possible to take into account all soil factors acting on it, but such an estimate is subject to errors related to the disadvantages of the methods used. To take into account heterogeneity of soils, it is recommended to take either many samples (Iwanek, 2005) or samples of large volume (Buchter et al., 1984). According to Novák et al. (2011), determining hydraulic conductivity requires 1 m³ soil samples. Neither many samples nor large samples, however, may be enough to avoid errors altogether. Even sampling itself and the transport of samples may cause laboratory errors in various ways (Fodor et al., 2011). Thus, taking undisturbed soil samples is a prerequisite for determining the impact of both rock fragments and soil macropores on hydraulic conductivity of forest stony soils.

This paper, thus, supplements the method for estimating hydraulic conductivity using undisturbed soil samples (Ilek and Kucza, 2014; Kucza and Ilek, 2016). We propose here how to enhance the method by taking into account the impact of rock fragments and soil macropores on the hydraulic conductivity of forest stony soils, and in particular on the hydraulic conductivity of the medium conducting water in the soil, that is, soil matrix.

2. Material and methods

2.1. Research area and sampling

We evaluated the proposed method on 33 undisturbed samples taken from the humus A (19 samples) and mineral B (14 samples) soil

layers, from selected stands of the Myślenice Forest District, located in south-central Poland. A detailed description of the sampling method and the location of the research area can be found in Ilek and Kucza (2014).

2.2. Methodological assumptions of the research

The laboratory method for determining hydraulic conductivity of forest stony soils under steady saturated conditions proposed by Ilek and Kucza (2014) uses undisturbed samples. Since soil samples can have irregular shape, depending on the sampling method, the method estimates hydraulic conductivity based on the following formula:

$$K = K' \cdot W_k \quad (1)$$

where K is the hydraulic conductivity (m·h⁻¹) that would have been obtained if the sample had the shape of a regular cylinder; K' is the approximate value of hydraulic conductivity (m·h⁻¹), obtained from direct measurements of undisturbed and irregularly shaped samples; and W_k is a conversion coefficient that takes into account the influence of irregular shape of the sample on the volume of water flowing through it.

In this work, in addition to taking undisturbed samples, we adopted an additional methodological assumption: When estimating saturated hydraulic conductivity, one should consider the hydraulic conductivity of soil matrix (< 2 mm), which is the medium conducting water in the soil. When calculating it, one should consider how rock fragments, soil macropores, and the irregular shape of a sample affect the volume of water flowing through the sample. Therefore, to calculate the hydraulic conductivity of soil matrix of a forest stony soil, we adopted the following formula:

$$K_{fe} = K'_{rc} \cdot W_r \cdot W_k \quad (2)$$

where K_{fe} is the hydraulic conductivity of soil matrix (m·h⁻¹); K'_{rc} the approximate value of hydraulic conductivity, reduced by the influence of soil macropore(s) (m·h⁻¹); and W_r is a reduction coefficient that takes into account the impact of rock fragments on hydraulic conductivity.

Since Ilek and Kucza (2014) and Kucza and Ilek (2016) described the influence of irregular shape of samples on hydraulic conductivity and the method of calculating the coefficient W_k , in the following sections we will describe only the method of measuring the coefficient K'_{rc} and the reduction coefficient W_r .

2.3. Hydraulic conductivity K'_{rc}

The first step in determining the K_{fe} coefficient for individual soil samples is to determine the approximate coefficient K' under steady saturated conditions, in accordance with the procedures presented by Ilek and Kucza (2014). The value of K' obtained from direct measurements on an undisturbed sample includes the influence of rock fragments, soil macropores, and the irregular shape of the sample on the volume of water flowing through the given sample. If, after determining K' , the sample contains at least one soil macropore, the next step is to determine the hydraulic conductivity K'_{rc} , a coefficient that takes into account the influence of the soil macropore(s).

During the research, we distinguished two main types of macropores:

1. Empty macropores (Fig. 1).
2. Macropores filled with a material with a reduced bulk density relative to the rest of the sample (Fig. 2a).

In the case of filled macropores, we removed their contents (Fig. 2b) and filled the resulting space with molten plasticine (Fig. 2c). The relatively small diameter and volume of empty macropores make it difficult to apply quick-drying molten plasticine. Therefore, they were



Fig. 1. Examples of empty macropores in soil samples.

usually filled with silicone, using a thin dispensing tip. Next, the sample was remeasured, and the coefficient K'_{rc} was recalculated at the cross-sectional area of the sample, which was reduced by the average cross-sectional area occupied in the sample by the medium filling the macropore (A_C). The area A_C (m^2) was calculated with the following formula:

$$A_C = \frac{V_C}{L} \quad (3)$$

where V_C is the volume of the medium filling the macropore, determined by the method of instantaneous displacement of water in a measuring cylinder (m^3); and L is the sample height (m).

The hydraulic conductivity of individual macropores (K_C) was determined from the difference between K' and K'_{rc} coefficients. In the case of samples with no soil macropores, K'_{rc} was equal to K' .

2.4. The conversion coefficient W_r

The conversion coefficient W_r , which takes into account the

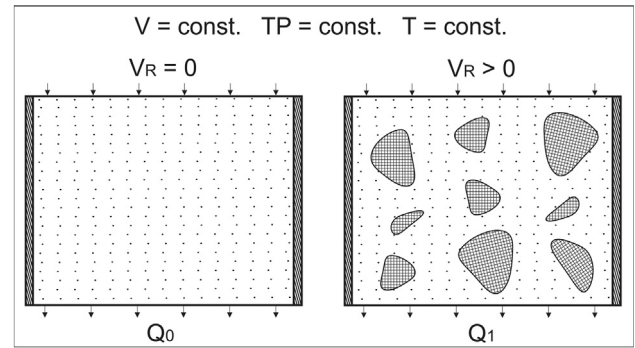


Fig. 3. The methodical assumptions of the experiment on the influence of the volume fraction of rock fragments on hydraulic conductivity. Explanations: Q_0 is the volume of water flowing through a sample of a homogeneous soil matrix, Q_1 is the volume of water flowing through a sample containing rock fragments, V is the volume of the sample, TP is the total porosity of soil matrix, T is the time of water flow measurement, and V_R is the volume of rock fragments.

influence of rock fragments on hydraulic conductivity of soil matrix was estimated with the following formula:

$$W_r = 1 - \frac{L_{RC}}{100} \quad (4)$$

where L_{RC} is the actual loss of water flow through soil matrix due to the presence of rock fragments with a given volume and average diameter in this medium (%).

To determine the W_r coefficient for individual undisturbed soil samples, we conducted experiments on artificial samples to analyse how the volume fraction of rock fragments (R_v) and their average diameter (D) affect L_{RC} (Figs. 3 and 4). We assumed that the volume of water flowing in a given time through a sample of a homogeneous soil matrix (with a given volume and porosity) can be related to a volume of water flowing through a soil matrix of the same porosity, but located in a sample containing a defined volume of rock fragments with a given diameter. These relationships are as follows:

$$Q_1 = Q_0 \cdot \left(\frac{V - V_R}{V} \right) \quad (5)$$

or

$$Q_1 \neq Q_0 \cdot \left(\frac{V - V_R}{V} \right) \quad (6)$$

where Q_1 is the volume of water flowing through the soil matrix of a sample that contains rock fragments ($cm^3 \cdot s^{-1}$); Q_0 is the volume of water flowing through a sample of a homogeneous soil matrix ($cm^3 \cdot s^{-1}$); V is a sample volume (cm^3); and V_R is a volume of rock fragments (cm^3).

If the equality (5) holds, (i) there is a balance between factors that increase and decrease water infiltration through the soil matrix because of rock fragments; (ii) rock fragments do not affect the volume of water



Fig. 2. The method of testing the influence of a soil macropores on hydraulic conductivity, where a – a soil macropore filled with a material of reduced volumetric density, b – an empty space created after the removal of the soil material filling the macropore, c – a space partially filled with molten plasticine.

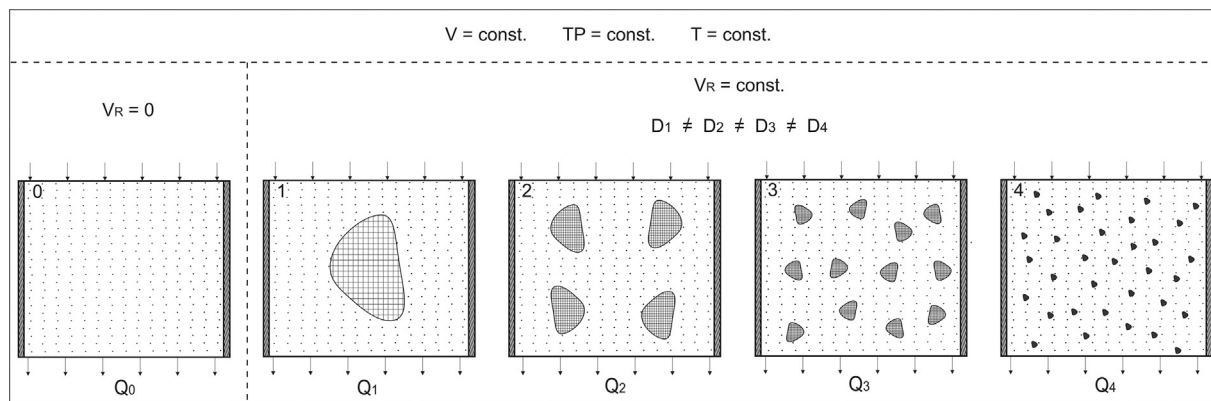


Fig. 4. The methodical assumptions of the experiment on the influence of the average diameter of rock fragments D on the volume of water flowing through a soil matrix of the samples containing rock fragments. Explanations: Q_0 is the volume of water flowing through a sample of a homogeneous soil matrix, Q_1 – Q_4 is the volume of water flowing through samples containing rock fragments of varying diameters (D_1 – D_4), V is the volume of the samples, V_R is the volume of rock fragments, TP is the total porosity of soil matrix, and T is the time of water flow measurement.

flowing through the soil matrix; and (iii) the differences between Q_0 and Q_1 are related only to the volume of rock fragments in the sample, which is why the percentage reduction in the volume of water flowing through the sample can be considered directly proportional to the volume fraction of rock fragments (R_V).

Under (6), so when the equality (5) does not hold, rock fragments affect the volume of water flowing through the soil matrix either positively (by enlarging water flow, e.g., through lacunar pores) or negatively (e.g., rock fragments with different size causing different curvatures of water flow pathways). Thus, the loss of water flow through a soil sample containing rock fragments (L_Q , %) is as follows:

$$L_Q = \left(\frac{Q_1 - Q_0}{Q_0} \right) \cdot 100 \quad (7)$$

while the actual impact of rock fragments on the decrease in water flow through a soil matrix is as follows:

$$L_{RC} = L_Q + R_V \quad (8)$$

where L_{RC} is the actual loss of water flow through soil matrix that is due to rock fragments (%); L_Q the loss of water flow through a soil sample containing rock fragments; and R_V is the volume fraction of rock fragments of the given diameter (%).

The experiments on the influence of rock fragments on loss of water flow through a soil sample containing rock fragments L_Q (Eq. (7)) and L_{RC} (Eq. (8)) was conducted using artificial samples, prepared in a cylinder with a diameter of 125 mm and a height of 60 mm. The soil matrix was calibrated sand with an average diameter of 0.17 mm. To ensure good contact between the cylinder walls and the side surface of a sample, and to prevent the formation of wall leakage during testing, the inner wall of the cylinder was covered with a thin layer of aerosol silicone and then a layer of sand. When the silicone dried, the cylinder was placed in the apparatus for testing the hydraulic conductivity of undisturbed samples (Ilek and Kucza, 2014).

The experiment started off by determining the hydraulic conductivity of five samples of pure sand. These samples were prepared by gradually pouring sand into the cylinder, while raising and lowering the water level in the outer cylinder of the apparatus. Such inducing of artificial variation in the water level resembled conditions of sand compaction and protected the samples against aeration. As a constant value of sand's hydraulic conductivity, we took the mean of five measurements (0.485 – $0.499 \text{ m}\cdot\text{h}^{-1}$), which was $0.492 \text{ m}\cdot\text{h}^{-1}$. The volume of water flow Q_0 that corresponded to this mean value was assumed as a constant used to estimate the influence of rock fragments on hydraulic conductivity. This effect was determined at six different volume fractions of rock fragments (R_V): 31.7, 21.2, 15.1, 10.3, 7.1, and 2.9%. For a given R_V , 7–8 replications were used, varying in the mean diameter of

rock fragments (D). These R_V fractions were determined based on the volume occupied in the sample by the largest rock fragment in a given series of the experiment, determined using the method of instantaneous displacement of water in a measuring cylinder. The very same volumes of rock fragments with smaller diameters D were measured in the same way in the experiments with the other values of R_V . For this, we used rock fragments of a roughly spherical shape, sampled from the research area (Magura sandstone). First, we used sieves of 2, 5, 10, 12, 25, 32, and 75 mm to divide the rock fragments into fractions, and then immersed them in water for 14 days. For individual R_V fractions, the rock fragments with the largest diameters were selected in the diameter range from 25 to 75 mm, and the mean diameter of rock fragments (D) in the sample was calculated based on the following formulas:

$$V_D = \frac{V_R}{n} \quad (9)$$

and

$$D = \frac{\sqrt[3]{V_D/(1.33 \cdot \pi)}}{5} \quad (10)$$

where V_D is the volume of an average rock fragment (cm^3); V_R is the total volume of rock fragments in a sample (cm^3); n is the number of rock fragments in the sample; D is the diameter of an individual rock fragment, equal to the average diameter of rock fragments in the sample (mm).

To study how rock fragments influence hydraulic conductivity, we conducted an experiment on 44 artificial samples. They were prepared under artificial fluctuations in the water level in the external cylinder of the apparatus, the rock fragments being distributed uniformly throughout the sample. This experiment led us to formulate the following relationship between the actual loss of water flow through soil matrix that is due to rock fragments (L_{RC} , %), the volume fraction of rock fragments (R_V , %), and their average diameter (D , mm):

$$L_{RC} = b_0 + b_1 \cdot R_V + b_2 \cdot R_V \cdot D + b_3 \cdot |(D - b_4)/(D + b_4)| \quad (11)$$

where b_0 , b_1 , b_2 , b_3 , and b_4 are non-linear regression coefficients, estimated using Statistica 13.1 PL software.

Derived based on the artificial samples, Eq. (11) was used to determine L_{RC} and W_r (Eq. (4)) for 33 undisturbed samples. To this aim, each sample was washed through a 2 mm sieve, and the rock fragments separated that way were then sieved through 5, 10, 12, 25, and 32 mm sieves. Next, for each sample, we determined the volume fraction of rock fragments of a given fraction, the total volume fraction of rock fragments (R_V), and their average diameter (D). The diameter D was calculated as the weighted mean volume of a given rock fragment fraction (measured by the method of instantaneous displacement of

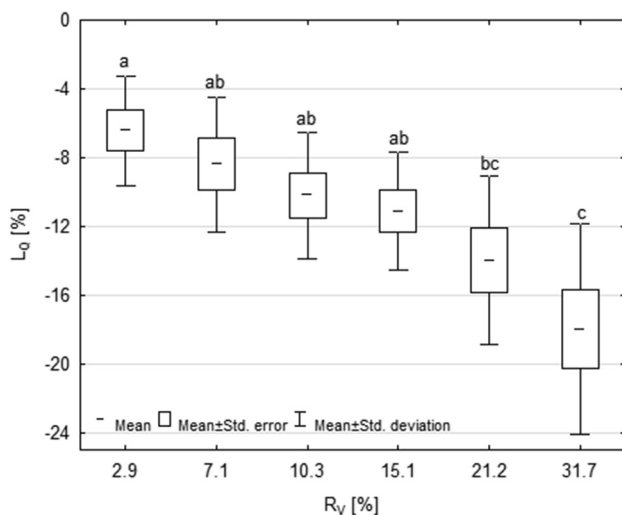


Fig. 5. The percent loss of water flow through a soil sample containing rock fragments (L_Q) against their volume fractions R_V . Different letters represent a significant difference between the corresponding R_V fractions in terms of L_Q , according to the *post-hoc* Tukey's HSD test at the significance level of 0.05.

water in a measuring cylinder) to the total volume of rock fragments in the sample. We estimated bulk density (BD), as dry mass of the sample per unit of volume; and soil matrix bulk density (BD_{fe}), as dry mass of the sample per unit of volume, excluding the volume of rock fragments. For samples containing filled macropores, we also estimated the bulk density of a macropore (BD_C), organic matter content in the material filling the macropore (OM_C), and organic matter content in the remaining part of the sample (OM). The organic matter contents were determined by the loss-on-ignition method, at a temperature of 550 °C.

3. Results

3.1. The influence of rock fragments on the loss of water flow rate

The percentage loss of water flow through samples containing rock fragments (L_Q) ranged from -24.8 to -1.6% , the mean loss increasing with the increase in rock fragment fraction R_V (Fig. 5) and reaching the maximum loss for R_V of 31.7%. The actual loss of water flow through soil matrix (L_{RC}) ranged from -5.0 to $+24.4\%$. Under R_V of 10.3%, the average L_{RC} loss was close to zero (Fig. 6), which may indicate a neutral influence of rock fragments on water infiltration through soil matrix. Negative L_{RC} losses under R_V below 10.3% indicate rock fragments' negative (inhibitory) effect. Only under R_V over 10.3%, rock fragments accelerate water flow through soil matrix and the largest positive L_{RC} values occurred under R_V of 31.7% (Fig. 6). In addition to the fraction of rock fragments, their mean diameter affected L_{RC} , too (Fig. 7). Regardless of the fraction, the highest positive effect on the hydraulic conductivity of soil matrix occurred under the smallest diameters of rock fragments. Regardless of the diameter D , under R_V of 15.1, 21.2, and 31.7%, rock fragments usually enhanced water infiltration through the soil matrix. When diameters D were close to 20 mm, the lowest impact was $+6.9\%$ under R_V of 31.7%; under R_V of 15.1 and 21.2%, the impact was below $+0.5\%$. Under R_V of 2.9, 7.1, and 10.3%, the positive influence of rock fragments on the water permeability of the soil matrix was observed only at the smallest diameters D , while the largest negative effect was observed under the 18–20 mm diameters, where the L_{RC} loss was in the range from -9.0 to -5.0% . Under the lowest R_V fractions, the neutral influence of rock fragments on the permeability of the soil matrix was observed for both the largest diameters and those close to 10 mm (Fig. 7). Table 1 shows the estimates, along with their statistical significance, of the parameters of Eq. (11), which represents the dependence of the actual loss of water flow

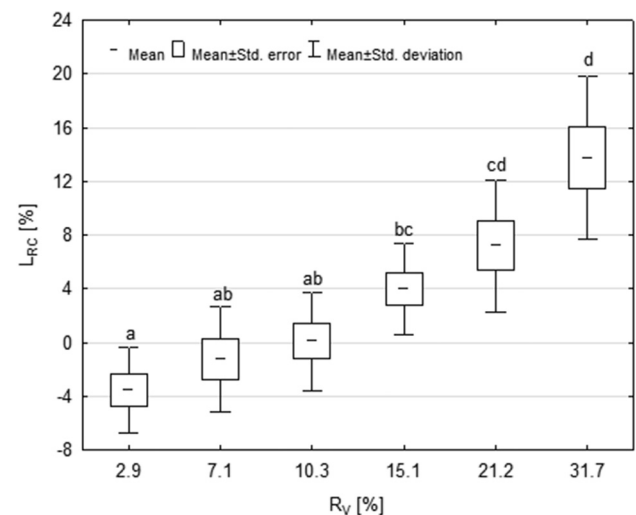


Fig. 6. The actual loss of water flow through soil matrix (L_{RC}) for different volume fractions of rock fragments (R_V). Different letters represent a significant difference between the corresponding R_V fractions in terms of L_{RC} , according to the *post-hoc* Tukey's HSD test at the significance level of 0.05.

through soil matrix (L_{RC}) on the volume fraction of rock fragments (R_V) and their average diameter (D). The model explained over 95% of the variability in L_{RC} .

3.2. Hydraulic conductivity of forest stony soils

Table 2 summarises various properties and other coefficients necessary to determine hydraulic conductivity of 33 samples and their soil matrix, obtained from the top layers of forest stony soils. The mean bulk density of samples containing rock fragments was $0.665 \pm 0.053 \text{ g}\cdot\text{cm}^{-3}$ (standard error of the mean) in layer A and $1.184 \pm 0.036 \text{ g}\cdot\text{cm}^{-3}$ in layer B. Soil matrix bulk density in layer A ranged from 0.217 to $0.856 \text{ g}\cdot\text{cm}^{-3}$ (mean of $0.592 \pm 0.048 \text{ g}\cdot\text{cm}^{-3}$), while of layer B from 0.859 to $1.298 \text{ g}\cdot\text{cm}^{-3}$ (mean of $1.095 \pm 0.034 \text{ g}\cdot\text{cm}^{-3}$). The volume fraction of rock fragments (R_V) in the samples tested ranged from 0.2 to 21.7%; in layer A, mean R_V was $4.7 \pm 1.1\%$, while in layer B $8.5 \pm 1.3\%$. The mean diameters of rock fragments (D) in layers A and B were similar: $11.6 \pm 1.0 \text{ mm}$ in layer A and $12.4 \pm 0.7 \text{ mm}$ in layer B. The actual loss of water flow through soil matrix (L_{RC}), estimated with the model (11), showed both positive and negative impacts on hydraulic conductivity (Table 2). In layer A, the mean L_{RC} loss was $-3.5 \pm 0.5\%$ while in layer B $-1.7 \pm 0.9\%$. The mean reduction coefficient W_r was 1.035 ± 0.005 in layer A and 1.017 ± 0.009 in layer B.

Among nine samples that contained soil macropores, six contained empty (Table 3) and three filled (Table 4) macropores. Most of the macropores, including all the filled ones, were found in layer A. Usually, filled macropores occupied a much larger volume in the samples tested than empty ones did, and their bulk density was on average $0.248 \text{ g}\cdot\text{cm}^{-3}$ lower than that of the rest of the sample (Table 4). The hydraulic conductivity of the empty macropores ranged from 0.161 to $3.354 \text{ m}\cdot\text{h}^{-1}$, while that of the filled macropores from 0.651 to $0.976 \text{ m}\cdot\text{h}^{-1}$, meaning that the soil macropores significantly increased the hydraulic conductivity of the samples (Fig. 8a). Empty macropores increased the K' value from two to nearly twenty-four times while filled macropores from three to five times.

The hydraulic conductivity K_{fe} – which takes into account the influence of macropores, rock fragments, and irregular sample shape – showed a clear relationship to the bulk density of soil matrix BD_{fe} (Fig. 8b), the highest K_{fe} values occurring under BD_{fe} below $0.300 \text{ g}\cdot\text{cm}^{-3}$. According to the Mann-Whitney U test, soil layers A and B differed in terms of the K_{fe} coefficient ($p < 0.0001$, $\alpha = 0.05$). The

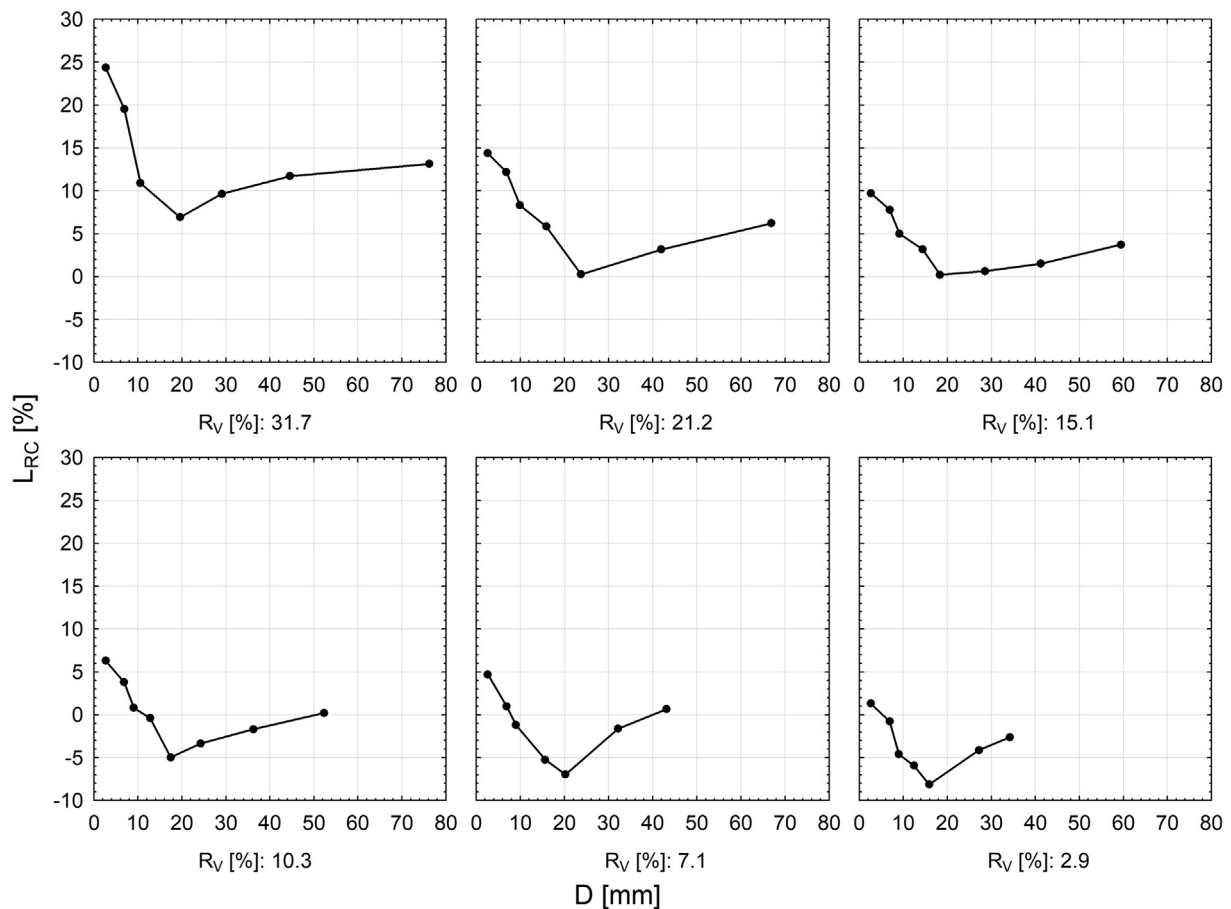


Fig. 7. Dependence of the actual loss of water flow through soil matrix due to the presence of rock fragments (L_{RC}) on their diameter (D) under varying volume fractions (R_V).

Table 1
Summary of the estimation of the nonlinear regression model (11).

Parameter	Estimate	Standard error	<i>t</i> statistics	<i>p</i> -value	R^2
b_0	−10.9	0.57	−19.2	< 0.001	0.954
b_1	0.70	0.03	20.8	< 0.001	
b_2	−0.004	0.001	−4.8	< 0.001	
b_3	15.4	1.08	14.2	< 0.001	
b_4	19.8	0.89	22.3	< 0.001	

mean hydraulic conductivity of soil matrix of layer A ($0.541 \pm 0.078 \text{ m}\cdot\text{h}^{-1}$) was almost three times higher than that of layer B ($0.191 \pm 0.031 \text{ m}\cdot\text{h}^{-1}$).

4. Discussion

Arising as a result of soil fauna activity and plant root death, macropores determine the main route of water infiltration in the soil and influence hydraulic conductivity (Mbagwu, 1995; Kodešová et al., 2006). Fig. 8a and Tables 3 and 4 confirm these claims. Léonard and Rajot (2001) showed that the activity of termites increases, by two or three times, water infiltration in the soil, and that the presence of

Table 2
Summary of the properties of undisturbed samples, obtained from the top layers of forest stony soils.

Variable	N	Mean	Std. error	Median	Min	Max	SD	CV (%)
BD [$\text{g}\cdot\text{cm}^{-3}$]	33	0.885	0.056	0.888	0.264	1.415	0.324	36.6
BD_{fe} [$\text{g}\cdot\text{cm}^{-3}$]	33	0.805	0.054	0.815	0.217	1.298	0.307	38.2
R_V [%]	33	6.3	0.9	5.0	0.2	21.7	5.1	80.3
D [mm]	33	11.9	0.6	12.7	3.5	19.9	3.6	30.4
L_{RC} [%]	33	−2.7	0.5	−2.7	−7.9	6.9	2.9	104.5
W_r	33	1.027	0.005	1.027	0.931	1.079	0.029	2.79
W_k	33	1.128	0.005	1.127	1.075	1.223	0.031	2.74
K' [$\text{m}\cdot\text{h}^{-1}$]	33	0.6346	0.1259	0.3694	0.0562	3.5019	0.7231	113.9
K_{fe} [$\text{m}\cdot\text{h}^{-1}$]	33	0.3926	0.0554	0.3729	0.0642	1.5728	0.3186	81.2

Where BD – bulk density, BD_{fe} – soil matrix bulk density, R_V – volume fraction of rock fragments, D – mean diameter of rock fragments, L_{RC} – the actual loss of water flow through soil matrix, determined with the formula (11), W_r – a reduction coefficient that takes into account the impact of rock fragments on hydraulic conductivity, W_k – a conversion coefficient that takes into account the irregular shape of a sample, K' – the approximate value of hydraulic conductivity, K_{fe} – the hydraulic conductivity of soil matrix that takes into account the influence of rock fragments, soil macropores, and irregular shape of samples, SD – standard deviation, CV – coefficient of variation.

Table 3
Parameters of the empty macropores observed in the samples.

No. of sample	Horizon	V_C [cm ³]	A_C [cm ²]	d [cm]	K_C [m·h ⁻¹]
1	A	38	12.6	0.40	0.311
3	B	28	7.1	0.30	0.161
11	A	59	19.6	0.50	2.168
12	B	95	15.9	0.45	0.708
17	B	80	20.0	0.51	3.354
31	B	64	15.9	0.45	0.662

Where V_C – the macropore's volume, A_C – the macropore's area, d – the macropore's mean diameter, K_C – the macropore's hydraulic conductivity.

Table 4
Parameters and properties of the filled macropores observed in the samples, compared to the properties of the remaining part of the corresponding sample.

No. of sample	Horizon	V_C [cm ³]	A_C [cm ²]	OM_C [%]	OM [%]	BD_C [g·cm ⁻³]	BD [g·cm ⁻³]	K_C [m·h ⁻¹]
6	A	140	46.7	17.7	11.5	0.518	0.717	0.976
18	A	100	33.3	12.7	9.4	0.681	0.856	0.651
29	A	85	28.3	15.3	12.2	0.324	0.693	0.775

Where V_C – a macropore's volume, A_C – a macropore's area, OM_C – organic matter content in the material filling the macropore, OM – organic matter content of the sample excluding the macropore, BD_C – bulk density of the material filling the macropore, BD – bulk density of the sample excluding the macropore, K_C – the macropore's hydraulic conductivity.

macropores of biological origin causes water to flow mainly through these macropores. Weiler and Naef (2003) found that during extreme rainfall, water flow is dominated by macropores formed by soil macrofauna (earthworms) – but the flow uses only some of the macropores, the remaining ones being either dry or reached by only a small amount of water. The quantification of a macropore effect requires two values of hydraulic conductivity: one that includes macropores and one that excludes them. Thus, reducing hydraulic conductivity by considering its dependence on soil macropore(s) seems to be justified.

The loss of water flow through a soil sample containing rock fragments (L_Q) suggests that rock fragments reduce the water flow rate in the soil (Fig. 5) and that the size of this reduction largely depends on

the volume fraction of rock fragments. Ma and Shao (2008) and Novák et al. (2011) also reported the decrease in infiltration due to the increase in rock fragment fraction. Yang et al. (2013) showed the greatest decrease in infiltration at the 50% fraction of rock fragments, and Mao et al. (2011) reported the lowest infiltration under the 20–30% fraction, above which it again increased. In Zhou et al. (2009), however, the lowest infiltration was under the 40% fraction. An increase in the rock fragment fraction in the soil, on the one hand, causes the area available for infiltrating water to decrease and the tortuosity of the flow to increase (Mehuys et al., 1975; Sauer and Logsdon, 2002). This increase in the rock fragment fraction, on the other hand, causes the bulk density of soil matrix (the water-conducting medium) to decrease, a process related to new pores being formed at the boundary between rock fragments and soil matrix (Poesen and Lavee, 1994; Torri et al., 1994).

Under R_V over 10%, water infiltration through the soil matrix accelerates (Fig. 6), which is probably related to the increase in the contact area between rock fragments and soil matrix. Under R_V over 10%, the influence of factors inhibiting the flow through the soil matrix is lower than that of the factors enhancing it. It is under R_V below 10% that the inhibiting factors gain dominance. A mean L_{RC} of 0.1% (± 1.3) obtained for samples containing 10.3% of rock fragments indicates that under this fraction the effects of factors inhibiting water flow through the soil matrix and the effects of factors enhancing it balance each other.

In addition to R_V , the mean diameter of rock fragments affects hydraulic conductivity, too. Diameters close to 20 mm have the least influence, but the effect may be positive, negative, or neutral, depending on R_V (Fig. 7). Diameters below 10 mm speed up water infiltration through the soil, increasing hydraulic conductivity up to about 25% under R_V of 31.7%. Beckers et al. (2016) showed that the saturated hydraulic conductivity of a loamy soil increases with the rock fragment fraction, but under some fractions the resistance to flowing water is greater for smaller stones than for larger ones. Yang et al. (2013) found an increase in hydraulic conductivity with an increasing rock fragment diameter, but Novák et al. (2011) reported an opposite effect – hydraulic conductivity decreased with an increasing rock fragment diameter.

Eq. (11), describing the dependence of actual loss of water flow through soil matrix on rock fragments and their average diameter, was developed using artificial samples, taking into account only one type of

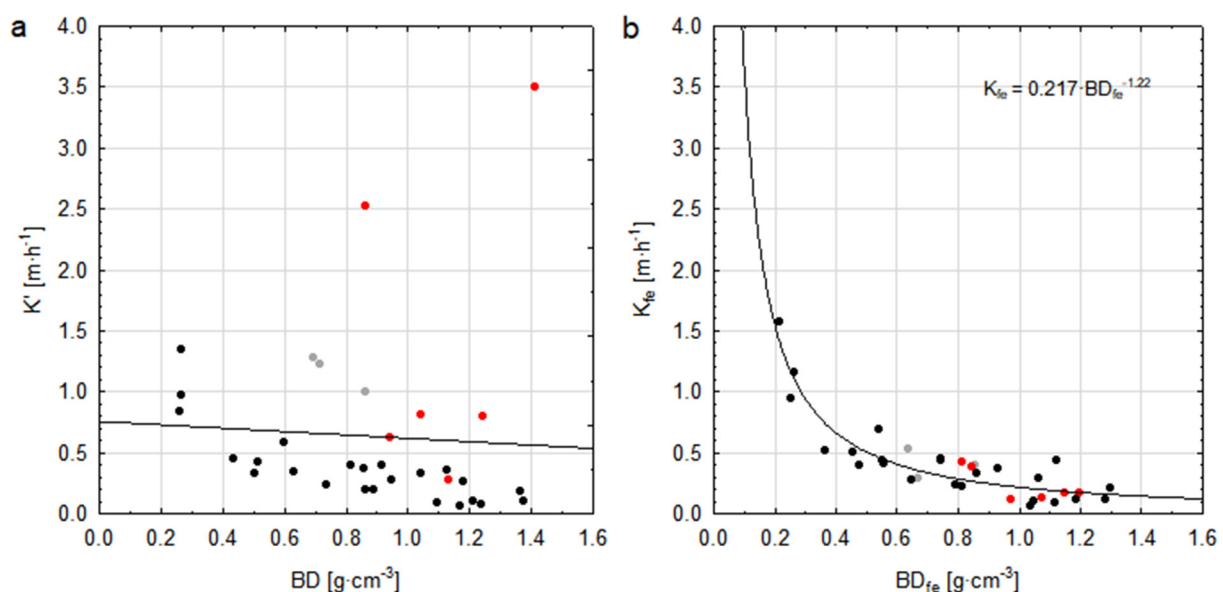


Fig. 8. Relationships between (a) the approximate hydraulic conductivity K' and the sample's bulk density BD , and (b) the hydraulic conductivity of soil matrix K_{fe} and soil matrix bulk density BD_{fe} . Red circles represent samples with empty soil macropores, grey circles samples with filled macropores, and black circles samples with no macropores. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

soil matrix: calibrated sand. The equation was then applied for undisturbed samples, characterised by natural particle packing and probably different hydraulic properties at the border between soil matrix and rock fragments than those of artificial samples. Direct contact between rock fragments and soil matrix leads to a change in porosity at the border between these two components (Grant and Struchtemeyer, 1959; Jung, 1960), causing local acceleration of water flow. Such a change can depend on differences in the bulk densities of rock fragments and soil matrix. For example, the processes of swelling and shrinkage of organic matter contained in forest soils may contribute to the formation of temporary voids along the soil/stone interface (lacunar pores) and cause a preferential flow, thereby increasing hydraulic conductivity (Hlaváčiková et al., 2016). Ma and Shao (2008) and Hlaváčiková et al. (2016) claimed that the shape, distribution, and orientation of rock fragments affect hydraulic conductivity. Therefore, to draw stronger conclusions about the effects of rock fragments on shaping the hydraulic conductivity of forest stony soils, future studies should consider the shape and position of rock fragments and use homogenised soil matrix of forest soil along with another, contrasting soil matrix (such as silt, clay and organic matter) to prepare different artificial samples.

5. Summary and conclusions

The present research on the effect of rock fragments on hydraulic conductivity in forest stony soils showed that rock fragments can either inhibit or accelerate water infiltration through a soil matrix, depending on the volume fraction and mean diameter of rock fragments. Further studies should analyse how the type of soil matrix and the shape and position of rock fragments affect hydraulic conductivity. We found that soil macropores play an important role in increasing the hydraulic conductivity of forest stony soils. Hydraulic conductivity strongly depends on soil matrix bulk density; it is thus justified to take into account the influence of soil macropores and rock fragments on hydraulic conductivity in laboratory tests.

The proposed method for sampling and sample testing as well as for testing and taking into account the influence of soil fragments and soil macropores on hydraulic conductivity has disadvantages: work, time consumption, and unsuitability for stony soils with large rock fragments (stones and boulders). These disadvantages, however, are outweighed by an important advantage: The method ensures the preservation of the natural structure of samples, as confirmed by the presence of empty and filled macropores in the collected samples. Thus, the method gives the opportunity to study the hydraulic conductivity of the macropores themselves and to reach reliable results for the hydraulic conductivity of heterogeneous soils with a relatively small sample size. Although in some situations a material filling macropores can be difficult to remove (e.g., when a filled macropore is located in the middle of an undisturbed sample), the method proposed gives the possibility to study biological activity and physical and chemical properties of macropores, which can be important not only for the field of hydrology, but also for other fields.

Conflict of interest

None declared.

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